

Network Capacity Assessment of Combined Heat and Power-Based Distributed Generation in Urban Energy Infrastructures

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Abstract—The growing penetration of distributed generation (DG) and use of combined heat and power (CHP) techniques have made electrical distribution systems and water distribution systems increasingly interdependent. This paper proposes an integrated dispatch model that offers a generalized means for determining the potential accommodated DG capacity along with appropriate DG sitting for given urban electricity and water system configurations. The AC optimal power flow (ACOPF)-based system model was developed for each phase of a modified electrical distribution test feeder. A corresponding water distribution system was then designed with its mathematical model developed. A mathematical model of the CHP-based DG system was also developed to couple the electricity and water systems. Results indicate the optimal DG capacity and its optimal sitting based on urban electricity and water infrastructures. The present water network cannot support the complete recovery of the exhaust heat. The mutual impacts on operational performances of CHP-based DG units and urban energy infrastructures were also investigated.

Index Terms—Combined heat and power (CHP), distributed generation (DG), load flow analysis, mathematical programming.

Sets

	NOMENCLATURE
B	Set of water network links (indexed by b).
C	Set of shunt capacitors (indexed by c).
G	Set of new generators (indexed by g).
L	Set of electrical lines (indexed by l).
N	Set of electrical buses (indexed by n).
P	Set of pumps (indexed by p).
V	Set of valves (indexed by v).
W	Set of water sources (indexed by w).
X	Set of external sources (indexed by x).

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Parameters

C^{HW}	Hazen-Williams roughness coefficient.
$(C, \rho)^W$	Water heat capacity and density.
$C_{c,n}^{(+,-)}$	(Max, Min) VAr capacity of c at n .
$(D, L)_b$	Link b diameter and length.
$D_n^{(P,Q)}$	(Real Power, Reactive Power) demand at n .
F_l^+	Max MVA capacity on l .
$(G, B)_L$	(Conductance, Susceptance) of l .
K^V	Energy loss coefficient.
$N_{(l,b)}^{(A,B)}$	(From, To) node of l and b .
$P_{g,n}^{(E,T),(+,-)}$	(Max, Min), (Electrical, Thermal) power output of g at n .
$(P, Q)_{x,n}^{X,(+,-)}$	(Max, Min), (P, Q) power output of x at n .
R_b^{Co}	Link b resistance coefficient.
T^A	Ambient temperature.
$T_{g,n}^{(SD,RT),(+,-)}$	(Max, Min), (Sending, Returning) water temperature of g at n .
$V_n^{(+,-)}$	(Max, Min) voltage at n .
W_n^D	Existing water demand at n .
$W_n^{H(+,-)}$	(Max, Min) water head pressure at n .
$W_{g,n}^{F(+,-)}$	(Max, Min) water flow of g at n .
$\beta_{(p,v)}^B$	Location of (p, v) corresponding to b .
$\beta_{(c,g,p,v,w,x)}^N$	Location of (c, g, p, v, w, x) corresponding to n .
η^{HE}	Heat exchanger efficiency.
$\phi_{(g,x)}$	Power angles of g and x .
Variables	
$c_{g,n}^{(WT,Wf)}$	Correction factor of (Water Temperature, Water Flow) of g at n .

$f_l^{(A,B),(P,Q)}$	(P, Q) injection onto l at (From, To) node.
$p_{g,n}^{(E,T)}$	(Electrical, Thermal) power output of g at n .
$p_{g,n}^{(FE,FT)}$	Final (Electrical, Thermal) power output of g at n .
$(p, q)_{x,n}^X$	(Real, Reactive) power output by x at n .
$r_{w,n}^W$	Water supplied by w at n .
$s_{c,n}^C$	VAr injection by c at n .
$t_{g,n}^{(SD,RT)}$	(Sending, Returning) water temperature of g at n .
(v_n, δ_n)	(Voltage level, voltage phase) at n .
$w_{g,n}^D$	Water demand by g at n .
w_b^F	Water flow in b .
w_n^H	Water head at n .

I. INTRODUCTION

ENGINEERING resilient and sustainable urban energy infrastructures is a prerequisite for continual urban growth, made challenging due to the interdependency of the confining physical and social environment. Currently, rapidly evolving environmental concerns and deregulated market forces are driving distributed generation (DG) penetration into distribution systems. This is changing the way that traditional centralized electric power is supplied based on transmission lines and substations. Additionally, DG technologies promise to be efficient, environmentally friendly, and dependable electric power sources [1]–[3].

One important application of DG technologies is the combined heat and power (CHP). To promote energy efficiency, many gas-fired DG units, e.g., microturbines, fuel cells, internal combustion (IC) engines, etc., can provide thermal energy by recovering the exhaust heat during electricity generation, which could be used to provide hot water, air conditioning, and district heat supply. The CHP technique has been mature in today's market for industrial and commercial applications. Meanwhile, it is also making inroads into residential areas, where the water supply from water distribution networks explicitly affects the captured DG thermal energy [4], [5].

A wide variety of mathematical optimization-based studies have been conducted in the literature on network DG capacity analysis, sitting, and sizing problems. Approaches based on AC optimal power flow (ACOPF) models, presented in [6]–[8], consider intrinsic nonlinearities of DG locations and capacity analysis, in which DG connection points are fixed and the maximum DG power injections are evaluated. In [9] and [10], analytical approaches are developed to solve DG sitting and sizing problems. Multiobjective-based optimization approaches are proposed for generation capacity planning in [11] and [12]. In [13], gradient and second-order methods are used to determine DG sitting based on loss minimization.

In the urban environment, electricity and water infrastructures have developed independently. However, with growing DG penetration and employment of CHP techniques, they are becoming more interconnected. A number of publications have

investigated the connections between electricity and water distribution systems in areas of microgrid related optimization, district heating and cooling, etc.. However, few studies have been conducted to investigate the network capacity of CHP-based DG in the context of urban electricity and water infrastructures. In reference [14], the integrated load flow method is used in a simple case study to examine the interconnected electric power and district heating networks. Specifically, a nine node microgrid feeder with only one location-fixed CHP unit and an eleven node district heating network are used in this study. In [9] and [10], an approach for integrated optimal power flow for energy infrastructures, e.g., electric power, gas and district heating, is presented and the energy hub concept is first introduced as the coupling point between different energy infrastructures. A three bus system and a corresponding three node natural gas system are also used to examine the optimal dispatch of multiple energy carriers. Finally, a network flow optimization model is developed in [15] for addressing the economic and physical integrity of the national electric power system by integrating the electric power, gas, coal, and water systems; this study, however, is focused on the transmission level rather than the distribution level.

Based on the above research, a clear need emerges for the development of a model architecture to provide planning guidance, which integrates electricity and water distribution systems with the coupling of the CHP-based DG units. This is especially valuable in urban energy planning. To the best knowledge of the authors of this paper, no general optimization technique regarding the network capacity assessment of CHP-based DG has been published based on relatively large distribution energy systems. It is the objective of this paper to introduce a generalized approach that investigates optimal DG capacity and sitting, and analyzes mutual impacts on operational performances of CHP-based DG units, and urban electricity and water infrastructures.

II. ELECTRICAL SYSTEM MODEL DEVELOPMENT

A. Electrical System Selection and Modification

The IEEE 123 node test feeder [16] was selected as the electrical system for this study, which is a typical radial distribution feeder, as shown in Fig. 1. Reasonable assumptions were made to change it to a looped system with improved reliability and increased DG capacity accommodation. By adding external AC power sources (ExGens) and connecting nodes 51 and 300, the original radial feeder is divided into several sections. Each section is one subfeeder connecting two ExGens with a switch at one side of the subfeeder. The modified looped system is still not a complete three-phase system. To make the result evaluation more accurate, the ACOPF formulation was applied to each phase. Consequently, the nodal DG output is represented by summing up nodal DG electrical output and thermal output of all three phases.

B. ACOPF Formulation

A group of CHP-based DG units were assumed to be placed at each node with the capacity from zero to a set value, which is multiple times of the base DG unit. The DG electric power output and thermal output are treated as continuous variables with both lower and upper capacity bounds. The lower capacity bounds were defaulted as zero.

1) *Objective Function:* The objective of the integrated electrical and water systems is to maximize the CHP-based DG output, including electrical output and recovered thermal

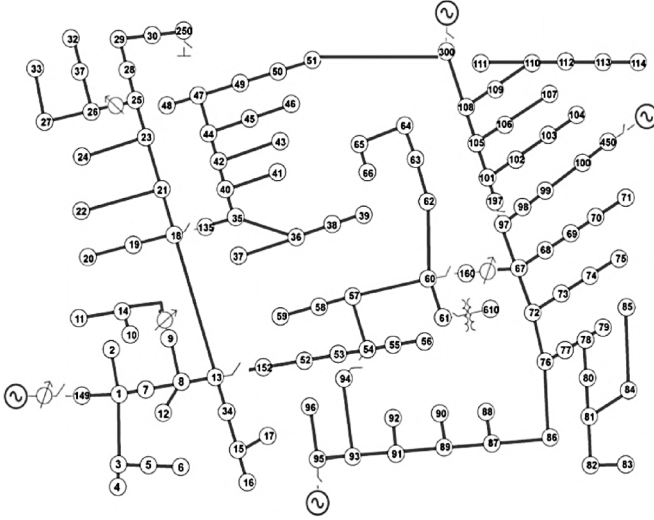


Fig. 1. Modified looped electrical system.

output. The electrical output is constrained by the electrical system model, while the recovered thermal output is constrained by electrical system model, CHP-based DG model, and water system model.

$$\max \sum_{g \in G} \sum_{n \in N} (p_{g,n}^{FE} + p_{g,n}^{FT}) \quad (1)$$

2) *Capacity Constraints of CHP-Based DG Units*: The actual max/min electrical and thermal capacity constraints of CHP-based DG units should be prorated with the total capacity changing of nodal DG units.

$$P_{g,n}^{(E,T),-} \leq p_{g,n}^{(E,T)} \leq P_{g,n}^{(E,T),+} \quad \forall g \in G \forall n \in N \quad (2)$$

3) *ExGens Constraints*: $\forall x \in X, \forall n \in N$:

$$(P, Q)_{x,n}^{X,-} \leq (p, q)_{x,n}^X \leq (P, Q)_{x,n}^{X,+} \quad \phi_x = 0 \quad (3)$$

4) *Nodal Voltage Level Constraint*: $\forall n \in N$:

$$V_n^- \leq v_n \leq V_n^+ \quad (4)$$

5) *Capacity Constraint of Shunt Capacitors*: $\forall c \in C$:

$$C_{c,n}^- \leq s_{c,n}^C \leq C_{c,n}^+ \quad (5)$$

6) *Real and Reactive Power Flow Constraints*: $\forall l \in L$:

$$\begin{aligned} f_l^{A,P} &= G_l v (N_l^A)^2 - v (N_l^A) v (N_l^B) \\ &\quad \times [G_l \cos(\delta(N_l^A) - \delta(N_l^B)) \\ &\quad + B_l \sin(\delta(N_l^A) - \delta(N_l^B))] \end{aligned} \quad (6)$$

$$\begin{aligned} f_l^{A,Q} &= B_l v (N_l^A)^2 - v (N_l^A) v (N_l^B) \\ &\quad \times [G_l \sin(\delta(N_l^A) - \delta(N_l^B)) \\ &\quad - B_l \cos(\delta(N_l^A) - \delta(N_l^B))] \end{aligned} \quad (7)$$

Apply the same real and reactive power flow [8] constraints to the other end of the line.

7) *Nodal Real and Reactive Power Balance Constraints*:

$$\sum_{\forall x \in X | \beta_x^N = n} p_{x,n}^X + \sum_{\forall g \in G | \beta_g^N = n} p_{g,n}^E$$

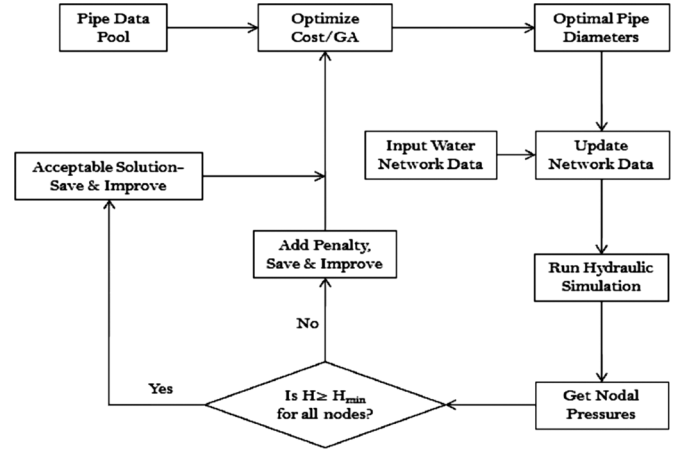


Fig. 2. Flowchart of the water distribution network design.

$$= D_n^P + \sum_{\forall l \in L | N_l^A = n} f_{l,n}^P + \sum_{\forall l \in L | N_l^B} f_{l,n}^P \quad (8)$$

$$\begin{aligned} &\sum_{\forall x \in X | \beta_x^N = n} q_{x,n}^X + \sum_{\forall g \in G | \beta_g^N = n} p_{g,n}^E \tan(\phi_g) + \sum_{\forall c \in C | \beta_c^N = n} s_{c,n}^C \\ &= D_n^Q + \sum_{\forall l \in L | N_l^A = n} f_{l,n}^Q + \sum_{\forall l \in L | N_l^B} f_{l,n}^Q \end{aligned} \quad (9)$$

8) *Thermal Constraints of Lines*: $\forall l \in L$:

$$\left(f_l^{(A,B),P} \right)^2 + \left(f_l^{(A,B),Q} \right)^2 \leq (F_l^+)^2 \quad \forall l \in L \quad (10)$$

III. WATER SYSTEM DESIGN AND MODEL DEVELOPMENT

In this section, according to the electrical system, a generalized urban water distribution system was designed and corresponding mathematical model was developed as well.

A. Water Network Design

In the water network, each node was assumed to represent a cluster of houses. The water demand was estimated based on the counted number of nodal households and the node loading information given by the electrical system.

Optimization techniques are often used to obtain a design of water distribution network. A genetic algorithm based optimization tool (GANetXL) [17] in combination with a hydraulic solver EPANET [18], is used to design a representative water distribution network by minimizing costs to choose appropriate pipe diameters. The flowchart of this design process is shown in Fig. 2.

The designed water distribution network corresponding to the electrical system is shown in Fig. 3. Each node in the water network corresponds to the same node in the electrical system. A reservoir is located in node 118 to provide all the water demand at each node. A pump is located between node 118 and node 250 to increase the water head up to the designed value (50 m) in node 250. This is to ensure that for all existing nodal water demand, the nodal water head is above the minimum required value. Two pressure reducing valves (PRV) are also included into this network to restrict the outflow.

B. Water System Model Development

To integrate the water system and electrical system, a mathematical model of the water system was developed based on the designed water distribution network.

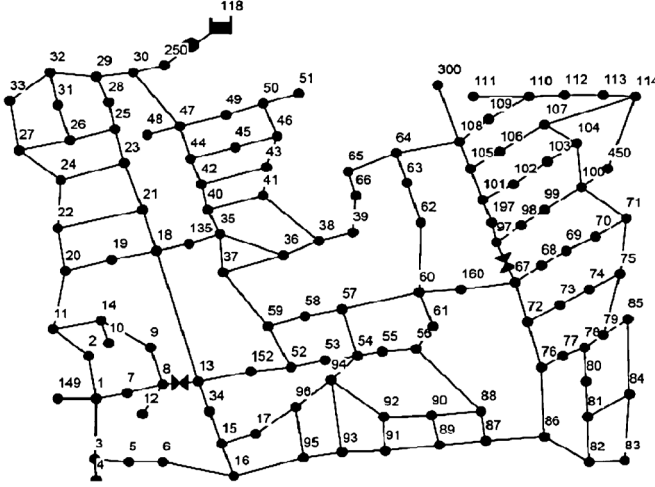


Fig. 3. Designed water distribution network.

1) *Pump Equation*: To represent the relationship between water head gain and water flow that a pump can deliver, one point (1140 m³/hr, 45 m of water head) on the pump curve was acquired to generate a representative pump (11) using EPANET.

$$w^H(p|\beta_p^N = n) - 5 = 60 - 0.00001154 \times [w^F(\beta_p^B)]^2 \quad (11)$$

The reservoir elevation was assumed as 5 m. The system was originally designed for a minimum pressure of 60 pounds per square inches (psi) or 42.2 m at all nodes during peak hours.

2) *Water Head Drop*: $\forall b \in B$: When water flows through pipelines, the head pressure drops due to friction, as described in (12) and (13) [19]. C^{HW} can be assumed as 130. The elevation of all nodes in the water network was assumed the same except the reservoir node such that the gravity impact can be neglected.

$$w^H(N_b^A) - w^H(N_b^B) = R_b^{Co} \times (w_b^F)^{1.85} \quad (12)$$

$$R_b^{Co} = 1.439 \times (C^{HW})^{-1.85} \times (D_b)^{-4.87} \times L_b \quad (13)$$

3) *Pressure Reducing Valves*: PRVs [19] are used in this model to restrict the water head at downstream node not exceeding 60 m. K^V for Globe type of PRVs was assumed as 5.

$$w^H(v|\beta_v^N = N_b^A) - w^H(v|\beta_v^N = N_b^B) = 6381.94 \times \frac{K^V \times w^F(\beta_v^B)^2}{D(\beta_v^B)^4} \quad \forall l = L \quad (14)$$

4) *Nodal Balance Equation* [19]: The water flows into a node equals the existing nodal water demand plus the water flows out of the node. The introduction of CHP-based DG units may result in additional nodal water demand.

$$\sum_{\forall w \in W|\beta_w^N = n} r_{w,n}^W - \sum_{\forall b \in B|N_b^A = n} w_b^F + \sum_{\forall b \in B|N_b^B = n} w_b^F = w_n^D + \sum_{\forall g \in G|\beta_g^N = n} w_{g,n}^D \quad (15)$$

5) *Nodal Water Head*: The system nodal water head is between 42.2 m and 50 m for all nodes during peak hours. The minimum water head may be changed for case studies.

$$W_n^{H-} \leq w_n^H \leq W_n^{H+} \quad \forall g \in G, \forall n \in N \quad (16)$$

IV. CHP-BASED DG MODEL

A. CHP-Based Microturbines

The CHP-based DG units are used as coupling points between electrical system and water system. The energy efficiency of microturbines can be boosted up from 30–40% to 85% with the exhaust heat recovered [20].

The exhaust heat from microturbines travels through the heat exchanger, normally the isolated or integrated part of microturbines. The pressurized water by the pump flows inside the heat exchanger, used as a medium to carry away part of the exhaust heat. The water flow operates as a looped circulating process and a certain amount of water is extracted from the water network periodically or in peak thermal demand besides the nodal existing water consumption. This additional water consumption by DG units is usually only a one-time use.

The microturbine C65 [21] was selected as a reference for CHP-based DG unit. The capacities of these series of products are available in a wide range, which can be modularized and show similar electrical and thermal characteristics as the selected reference unit.

B. Mathematical Formulation

The mathematical formulation takes into account impacts of units' electrical output, water flow rate, sending and returning water temperatures, and ambient temperatures:

1) *Water Temperature Constraints*: The minimum and maximum sending and returning water temperatures are 60 °F and 85 °F, and 220 °F and 185 °F, respectively [22].

$$T_{g,n}^{(SD,RT),-} \leq t_{g,n}^{(SD,RT)} \leq T_{g,n}^{(SD,RT),+} \quad \forall g \in G, \forall n \in N \quad (17)$$

2) *Water Consumption Constraint*: For single microturbine unit, the allowable water flow ranges between 5.68 m³/hr and 13.6 m³/hr [22]. The actual max/min nodal water consumption constraints of microturbines should be prorated with the total capacity changing of nodal DG units.

$$W_{g,n}^{F-} \leq w_{g,n}^D \leq W_{g,n}^{F+} \quad \forall g \in G, \forall n \in N \quad (18)$$

3) *Ambient Temperature Constraints*: $\forall g \in G, \forall n \in N$: When T_A exceeds 70 °F, both DG electrical output and thermal output decrease with T_A increasing, as shown in (20) and (19); when T_A is less than 70 °F, the electrical output keeps constant while the thermal output increases with T_A increasing, as shown in (22) and (21) [22]. The rating electrical capacity and thermal capacity of a single microturbine unit is 65 kW and 116.5 kW, respectively.

$$p_{g,n}^T \leq \frac{p_{g,n}^E}{65} \times [116.5 - 0.1905 \times (T_A - 70)] \quad (19)$$

$$p_{g,n}^{FE} = p_{g,n}^E \times [1 - 0.00577 \times (T_A - 70)] \quad (20)$$

$$p_{g,n}^T \leq \frac{p_{g,n}^E}{65} \times [116.5 - 0.4308 \times (70 - T_A)] \quad (21)$$

$$p_{g,n}^{FE} = p_{g,n}^E \quad (22)$$

4) *Heat Transfer Equation*: $\forall g \in G, \forall n \in N$: Heat energy is transferred from DG exhaust output to heat exchanger and carried away by the flowing water. The heat power transferred also depends on the water mass flow and temperature difference [23]. η^{HE} was assumed as 0.9.

$$\eta^{HE} \times p_{g,n}^T = C^W \times \rho^W \times w_{g,n}^D \times (t_{g,n}^{SD} - t_{g,n}^{RT}) / 3600 \quad (23)$$

5) *Correction Factors Constraints*: $\forall g \in G, \forall n \in N$: The inlet water temperature and water flow rate impact the recovered thermal output of microturbines, and thus correction factors of thermal output are generated [22]:

$$cf_{g,n}^{WT} = 1 - (t_{g,n}^{RT} - 140) \times 0.0023 \quad (24)$$

$$cf_{g,n}^{WF} = 0.83 + 0.02464 \times w_{g,n}^D - 0.0001496 \times (w_{g,n}^D)^2 \quad (25)$$

6) *Final DG Recovered Thermal Output*:

$$p_{g,n}^{FT} = p_{g,n}^T \times \eta^{HE} \times cf_{g,n}^{WT} \times cf_{g,n}^{WF} \quad \forall g \in G, \forall n \in N \quad (26)$$

V. PROBLEM SOLVING, CASE STUDIES, AND DISCUSSION

A. Implementation

The integrated system model was coded and implemented as a nonlinear mathematical programming (MP) problem in the AIMMS optimization modeling environment [24]. The MP was generated by AIMMS from the model structure and data, and then passed to the selected efficient commercial solver—the CONOPT 3.14V generalized reduced gradient solver. In its default settings, CONOPT 3.14V has proven to be competitive with the KNITRO interior point solver; it has shown reliability and efficiency in convergence [25]. Normally it is not guaranteed for nonlinear solvers to find the global optimum. The returned solution most often is a local optimum, or possibly no feasible solution can be found. The global solver such as Baron is time-consuming and usually only works well on relatively small NLP problems. In addition to the selected nonlinear solver, the nonlinear presolver and the multistart procedure [26] were also used to reduce problem size, search space, and increase chances of finding the global optimum by comparing and tracking all found feasible solutions.

The integrated system solving flowchart is shown in Fig. 4. The electrical system model was optimized first for each single phase to find the maximum DG electrical output. The resultant nodal DG electrical output is then used as the input parameters of upper bounds for the thermal constraints, as shown in (19) and (21). The optimal DG thermal output is achieved simultaneously when the electrical system model is optimized. However, as for the maximum DG recovered thermal output, it is not only constrained by the electrical system model, but also by the water system model and CHP-based DG system model. If the minimum nodal water pressure is violated, then less water will be extracted by DG units from water network to carry away the thermal energy from exhaust heat. Subsequently, the DG recovered thermal output is reduced accordingly. This iteration continues until no nodal water pressure violation occurs.

B. Case Studies

The electrical system model, water system model, and CHP-based DG model were developed individually and integrated together to constrain the CHP-based DG output. Multiple factors primarily affect the CHP-based DG electrical output and recovered thermal output, such as units' power factor (PF), capacity bounds, ambient temperature, and nodal water head.

1) *Capacity Bounds*: By changing the capacity of DG units to 0.065 MW, 0.13 MW, 0.325 MW, 0.65 MW, and 1 MW, five cases from Case 1 to Case 5 were studied to investigate the impact of capacity changes on DG units' performance and distribution. A fixed leading PF of 0.95 was assumed for all DG units. The electrical system is 4.16 kV, with real and reactive power load of 3.49 MW and 1.92 MVAR, respectively. The real and reactive power capacities of ExGens were assumed as ± 60

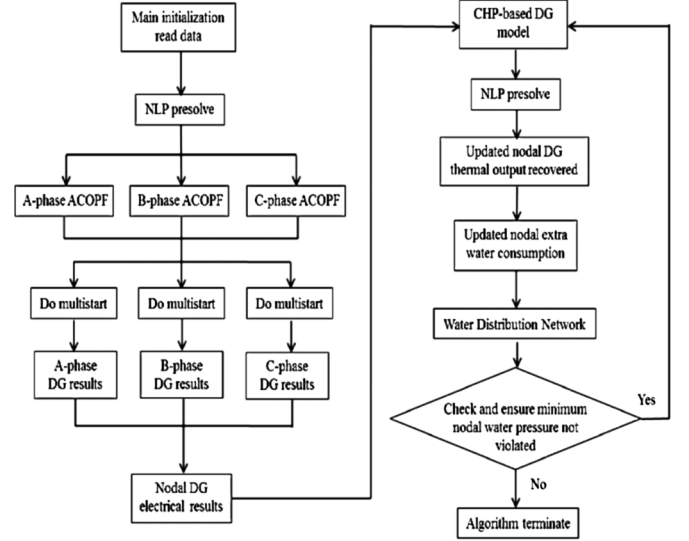


Fig. 4. Integrated system solving flowchart.

TABLE I
ELECTRICAL OUTPUT SUMMARY

Case Studies	Case 1	Case 2	Case 3	Case 4	Case 5
Total Electrical Output (MW)	16.51	22.37	24.49	27.78	31.3
Penetration Level (%)	100	87	68	62	45
Grids Real Power Output (MW)	-12.58	-18.39	-20.47	-23.73	-27.21
Grids Reactive Power Output (MVAR)	-2.47	-4.41	-4.96	-5.79	-6.98
Money Earned (\$)	3775	5517	6141	7118	8164
Money Earned per Household (\$)	2.16	3.16	3.52	4.08	4.68

MW and ± 50 MVAR, respectively. For capacity bound analysis, only the electrical system model was optimized, while the water and CHP based DG system models were not integrated.

The electrical output summary for this study is shown in Table I. The total DG electrical output increases with the rise in capacity bounds. The extra real and reactive power generated by DG units is sent back to grids. Assumptions were made that there were three peak hours and credits were given for the power sold back to grids at the rate of 10 cents for each kWh of electricity. The total and average money earned for each case appears as shown in Table I. Moreover for each case, it was also found that the reactive power output of shunt capacitors was zero, such that all the required reactive power was supplied by the DG units.

The penetration level refers to the ratio between the number of nodes connected with DG units and the total number of nodes. From Table I, it is found that not every node is appropriate to place a CHP-based DG unit and the penetration level reduces with the increase of capacity bounds. The reason for this is that with an increasing in the capacity bound, a node with DG units connected continues raising power output until a violation appears at that node, assuming no other constraints exist. The penetration level should be neither too high nor too low. This is because if the penetration level is too high, for e.g., in Case 1, where DG units are connected at every node, then the fault level and stability concerns grow significantly in this distribution system when all the units work at the same time. If the penetration level is too low, such as in Case 5 or even lower in other

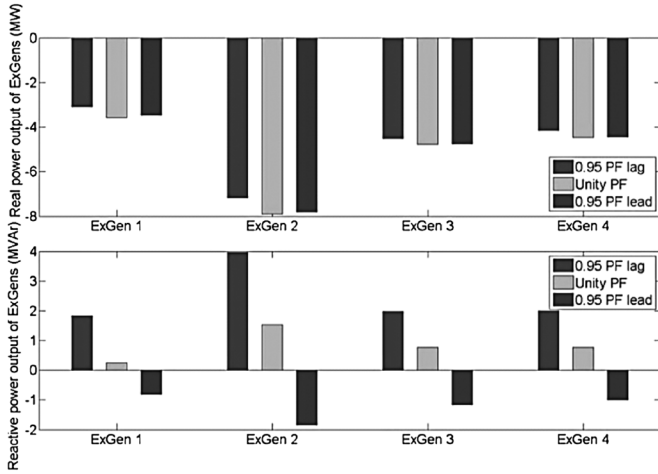


Fig. 5. Power output of ExGens at various PFs.

unstudied cases, the economic benefits will not be sufficiently achieved. This is because that building new thermal pipelines is costly if the load center is located far away from the thermal source. Therefore, in the following sections, Case 3 will be used as a basis to investigate the impacts of other factors.

2) *PF*: For power factor analysis, three different PFs were studied: 0.95 lag, unity, and 0.95 lead. Only the electrical system model of Case 3 (0.325 MW) was optimized for these three different PFs. The water and CHP based DG system models were not integrated. All other parameter settings are the same as those in the capacity bound analysis.

The results indicate that the maximum total DG electrical power output is 25.21 MW at the unity PF, which exceeds that at a 0.95 PF lag by 1.75 MW, and that at a 0.95 PF lead by 0.72 MW, respectively.

In Fig. 5, the real and the reactive power output of ExGens are shown that take into consideration different PFs. All of these ExGens absorb the extra real power delivered by the DG units, in which the most real power is sent to ExGen2, while ExGen1 absorbs the least real power. Similarly, for each ExGen, the maximum power absorbed always appears at the unity PF. All ExGens send reactive power to the system at PFs of 0.95 lag and unity, in which ExGen2 sends the most. When the PF is 0.95 lead, all ExGens absorb the extra reactive power from DG units. It is also observed that all shunt capacitors work in full capacity when the PFs are 0.95 lag and unity.

3) *Ambient Temperature*: For ambient temperature analysis, several different ambient temperatures, -4°F , 32°F , 60°F , 70°F , 80°F , 100°F , and 122°F , were used as references to study their influences on DG units' performance. For this case, the electrical, water and CHP-based DG system models were integrated together to maximize DG electrical output and recovered thermal output. For electrical system model, Case 3 (0.325 MW) and a unity PF were selected, and all other electrical parameter settings keep the same. The parameter settings of CHP-based DG system model have been shown in Section IV. The nodal water head in the water network was set as 40 m.

The total DG electrical output and recovered thermal output are shown in Fig. 6 with the changing ambient temperature. When the ambient temperature is below 70°F , the electrical output is in equilibrium and not affected, while the recovered thermal output decreases with the reduction of ambient temperature. When the ambient temperature is above 70°F , for every 10°F increase the electrical output and the recovered thermal

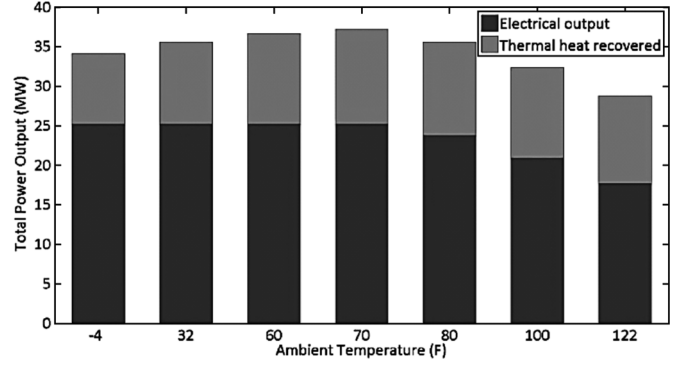


Fig. 6. Total DG output with changing ambient temperature.

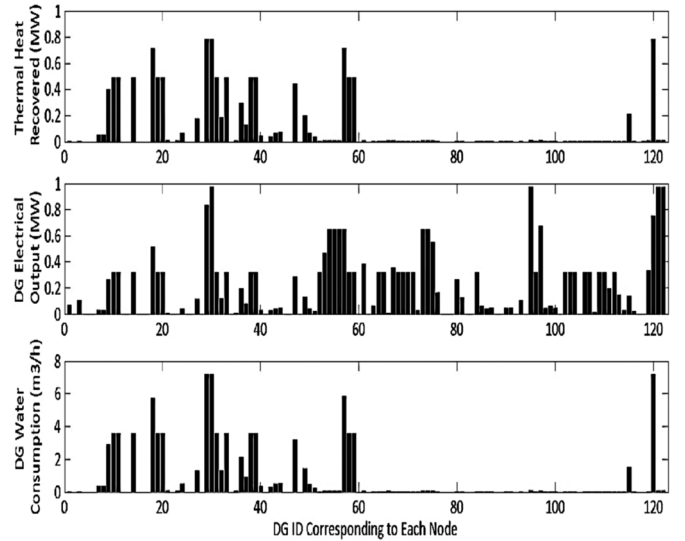


Fig. 7. Nodal power output and water consumption of the CHP-based microturbines.

output decreases 5.78% and 1.39%, respectively. The optimal ambient temperature is found at 70°F , at which the total DG power output reaches the maximum value of 37.24 MW including 12.03 MW of recovered thermal output and 25.21 MW of electrical output.

The nodal recovered thermal output, electrical output and the water consumption of the CHP-based DG units are shown in Fig. 7 with the ambient temperature of 70°F . It demonstrates that for each node with the electrical output, the thermal output may not be recovered at the same node due to the nodal water head and water flow rate. For DG IDs between 60 and 120, they correspond to the nodes with low water heads, as they are located at the far right side of the pump in the water distribution network. The water head automatically drops with increasing distance from the pump. If additional pumps are added to the right side of the water network, the water head will rise and more CHP-based microturbines in these nodes will become capable of recovering the exhaust heat. It is also found in Fig. 7 that wherever there is recovered thermal output, the water consumption is found in the same node.

4) *Nodal Water Head*: Nodal water head is a main factor constraining water consumption and recovered thermal output of DG units. It was checked to examine if the existing water distribution system could support the large penetration of CHP-based DG units. For nodal water head analysis, the electrical, water and CHP-based DG system models were integrated together to maximize DG output. For electrical system model, Case 3

TABLE II
POWER OUTPUT WITH DIFFERENT NODAL WATER HEAD LIMITS

Nodal Water Head (m)	36	38	40	42
Electrical Output (MW)	25.21	25.21	25.21	25.21
Recovered Thermal Output (MW)	23.70	18.62	12.03	4.21
Total Output (MW)	48.91	43.83	37.24	29.42

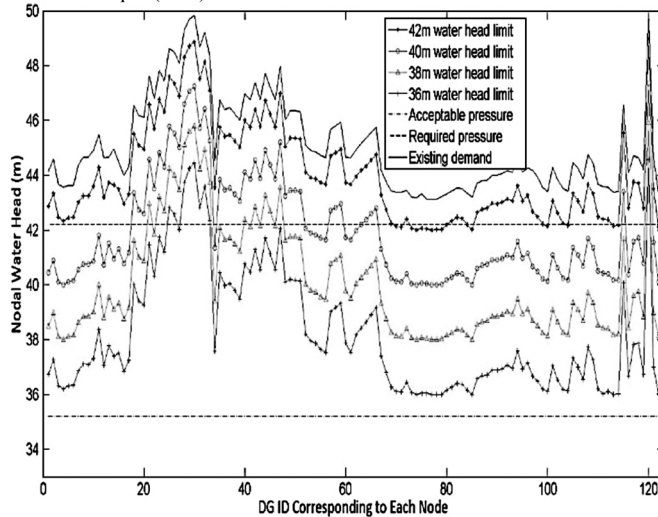


Fig. 8. Nodal water head with different water head limits.

(0.325 MW) and a unity PF were selected, and all other electrical parameter settings keep unchanged. The parameter settings of CHP-based DG system model are shown in Section IV. The ambient temperature of 70 °F was selected from the above analysis and discussion. For the water network, several different nodal water heads, 36 m, 38 m, 40 m, and 42 m, were selected to investigate their impacts on DG output. The required water head was set as 42.2 m and the acceptable water head was set as 35.2 m.

The total DG electrical output and recovered thermal output are shown in Table II. It is found that for all nodal water heads, electrical output remains unchanged, while the recovered thermal output decreases significantly with an increase in the nodal water head limit.

The nodal water head with different limit requirements is shown in Fig. 8. The top black line indicates the nodal water head with existing water demand, while all remaining curves represent nodal water heads with the extra water used by DG units under different water head limits. For DG IDs between 20 and 70, the nodal water heads are higher than all remaining nodes, with the exception of nodes clustered around 120. This is because these nodes are located close to the pump. It is also found that the more water extracted from the water network, the lower the nodal water head. For DG water consumption, the higher the nodal water head limit, the less the extra water available for recovering the exhaust heat. The CHP-based DG penetration increases with the reduction in the nodal water head limit. If an acceptable pressure drop of 10 psi (from 60 psi) is allowed, the water system is functional at an acceptable level during peak hours for the additional water demand used by the DG units.

The recovered thermal output is summarized in Table III; it is found that the total hourly hot water supply decreases significantly with an increase in the nodal water head limit. Similarly, the study time period was also assumed as three peak hours. As shown in Table III, the more thermal energy supplied, the more money is saved. Savings accrue by directly using the hot water

 TABLE III
RECOVERED THERMAL OUTPUT SUMMARY

Nodal Water Head Limit (m)	36	38	40	42
Total Hourly Hot Water Supply (m ³ /hr)	187.013	142.776	93.195	35.137
Total Thermal Supply (kWh)	71100	53700	36000	12630
Total Money Saved (\$)	7110	5370	3600	1263
Money Saved Per Household (\$)	4.074	3.077	2.063	0.724

supplied by the CHP-based DG units, which would otherwise have been bought from grids.

C. Discussion

The integrated mathematical model was solved as a nonlinear optimization problem. A local returned optimum may not definitely represent the global optimum. Additionally, it is also impractical and not guaranteed to find a global optimum for any large nonlinear planning problems. The algorithms of nonlinear presolve and multistart were employed and only a local solution was found, which proved to be the best results generated and are likely to be the global results. For this specific planning problem, a technique with an achievable solution provides more value than a global optimum which cannot be guaranteed for other cases [6]. It should be noted that in a distribution system without many meshed loops, such as that shown in Fig. 1, it is highly possible that a single local optimum exists.

It has been demonstrated that for all the cases studied: extra electrical output from DG units is available besides that used to meet the system load; however, the CHP application cannot be employed at every node and the existing water distribution network cannot support the complete recovery of exhaust heat. The performance of the integrated system depends on multiple factors. For real deployment, other concerns, e.g., investment, income, and location availability, should also be taken into account to size DG units. In reality, the PFs of most gas-fired DG units can be specified according to customer requirements. Furthermore, the adjustable indoor or outdoor temperature is suggested to provide an, ensuring that the gas-fired DG units are operated in optimal climatic conditions. Finally, the determination of water head limit during the planning period has to take into account present water demand, additional demand and potential thermal applications.

VI. CONCLUSIONS

The methodology developed for the integrated mathematical model is novel as this is the first instance of assessing the CHP-based DG network capacity by coupling generalized electrical distribution system, water distribution system and gas-fired CHP-based DG system. The problem of this DG capacity maximization is different from cost minimization problems as more congestion may appear not only in the electrical network, but also in the water network. The proposed method is generic and can also be used in a radial distribution system, as shown in [5]. The CHP-based DG units are placed at every node, rather than being fixed into some nodes, and function as continuous decision variables. This solution approach is new and being first employed in the DG network capacity assessment based on urban energy infrastructures.

REFERENCES

- [1] H. Nikkhajoei and R. H. Lasseter, "Distributed generation interface to the CERTS microgrid," *IEEE Trans. Power Del.*, vol. 24, no. 3, pp. 1598–1608, Jul. 2009.

- [2] P. Chiradeja and R. Ramakumar, "An approach to quantify the technical benefits of distributed generation," *IEEE Trans. Energy Convers.*, vol. 19, no. 4, pp. 764–773, Dec. 2004.
- [3] X. Zhang, G. Karady, and Y. Guan, "Design methods investigation for residential microgrid infrastructure," *Eur. Trans. Electr. Power*, vol. 21, pp. 2125–2141, 2011.
- [4] A. D. Peacock and M. Newborough, "Impact of micro-combined heat-and-power systems on energy flow in the UK electricity supply industry," *Energy*, vol. 31, pp. 1804–1818, 2006.
- [5] X. Zhang, G. Karady, K. Piratla, and S. Ariaratnam, "Nexus between distributed generation and urban water infrastructure," in *Proc. IEEE Power Eng. Soc. Gen. Meet.*, 2012, pp. 1–8.
- [6] C. J. Dent, L. F. Ochoa, and G. P. Harrison, "Network distributed generation capacity analysis using opf with voltage step constraints," *IEEE Trans. Power Syst.*, vol. 25, no. 1, pp. 296–304, 2010.
- [7] A. Keane and M. O'Malley, "Optimal allocation of embedded generation on distribution networks," *IEEE Trans. Power Syst.*, vol. 20, no. 3, pp. 1640–1646, Aug. 2005.
- [8] C. J. Dent, L. F. Ochoa, G. P. Harrison, and J. W. Bialek, "Efficient secure ac opf for network generation capacity assessment," *IEEE Trans. Power Syst.*, vol. 25, no. 1, pp. 575–583, 2010.
- [9] M. Geidl and G. Andersson, "Optimal power flow of multiple energy carriers," *IEEE Trans. Power Syst.*, vol. 22, no. 1, pp. 145–155, Feb. 2007.
- [10] M. Arnold, R. Negenborn, G. Andersson, and B. De Schutter, "Model-based predictive control applied to multi-carrier energy systems," in *Proc. IEEE Power Eng. Soc. Gen. Meet.*, 2009, pp. 1–8.
- [11] X. Zhang, R. Sharma, and Y. He, "Optimal energy management of a rural microgrid system using multi-objective optimization," in *Proc. IEEE PES Innov. Smart Grid Technol. Conf.*, 2012, pp. 1–8.
- [12] B. Kuri, M. Redfern, and F. Li, "Optimization of rating and positioning of dispersed generation with minimum network disruption," in *Proc. IEEE Power Eng. Soc. Gen. Meet.*, Jun. 2004, vol. 2, pp. 2074–2078.
- [13] N. S. Rau and Y. H. Wan, "Optimum location of resources in distributed planning," *IEEE Trans. Power Syst.*, vol. 9, no. 4, pp. 2014–2020, 1994.
- [14] M. Rees, J. Wu, and B. Awad, "Steady state flow analysis for integrated urban heat and power distribution networks," in *Proc. 44th Int. Univ. Power Eng. Conf.*, 2009, pp. 1–5.
- [15] E. M. Gil, "Modeling integrated energy transportation networks for analysis of economic efficiency and network interdependencies," in *Proc. North Amer. Power Symp.*, Rolla, MO, USA, 2003.
- [16] W. H. Kersting, "Radial distribution test feeder," *IEEE Trans. Power Syst.*, vol. 6, no. 3, pp. 975–985, 1991.
- [17] D. Savić, J. Bicik, and M. Morley, "A dss generator for multiobjective optimisation of spreadsheet-based models," *Environ. Model. Softw.*, vol. 26, no. 5, pp. 551–561, 2011.
- [18] L. A. Rossman, EPANET: User's Manual U.S. EPA, Cincinnati, OH USA.
- [19] L. W. Mays, *Hydraulic Design Handbook*. New York: McGraw-Hill, 1999.
- [20] Microturbines [Online]. Available: <http://www.energy.ca.gov/distgen/equipment/microturbines/microturbines.html>
- [21] C65 & C65 ICHP MicroTurbine Natural Gas [Online]. Available: http://www.capstoneturbine.com/_docs/datasheets/C65%20&%20C65-ICHP%20NatGas_331035E_lowres.pdf
- [22] Technical Reference, Application Guide—Model C65 Integrated CHP (ICHP) Capstone Turbine Corporation. Chatsworth, CA, USA.
- [23] B. Awad, M. Chaudry, J. Wu, and N. Jenkins, "Integrated optimal power flow for electric power and heat in a microgrid," in *Proc. 20th Int. Conf. Electr. Distrib.*, Jun. 2009.
- [24] J. Bisschop and M. Roelofs, AIMMS—The User's Guide Paragon Decision Technology, 2006.
- [25] Full Details on AIMMS and Associated Solvers [Online]. Available: <http://www.aimms.com>
- [26] J. Bisschop and M. Roelofs, AIMMS—The Language Reference Paragon Decision Technology, 2006.

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